

Green Life: Identify Module

Instruction before accessing the system:

1. Install C++ Redistributable

- ✓ <https://www.tensorflow.org/install>

Install TensorFlow 2

TensorFlow is tested and supported on the following 64-bit systems:

- Python 3.8–3.11
- Ubuntu 16.04 or later
- Windows 7 or later (with C++ redistributable)
- macOS 10.12.6 (Sierra) or later (no GPU support)
- WSL2 via Windows 10 19044 or higher including GPUs (Experimental)

- ✓ Install package of X64

Architecture	Link	Notes
ARM64	https://aka.ms/vs/17/release/vc_redist.arm64.exe	Permalink for latest supported ARM64 version
X86	https://aka.ms/vs/17/release/vc_redist.x86.exe	Permalink for latest supported x86 version
X64	https://aka.ms/vs/17/release/vc_redist.x64.exe	Permalink for latest supported x64 version. The X64 Redistributable package contains both ARM64 and X64 binaries. This package makes it easy to install required Visual C++ ARM64 binaries when the X64 Redistributable is installed on an ARM64 device.

2. Install Miniconda

- ✓ <https://www.tensorflow.org/install/pip#windows-native>

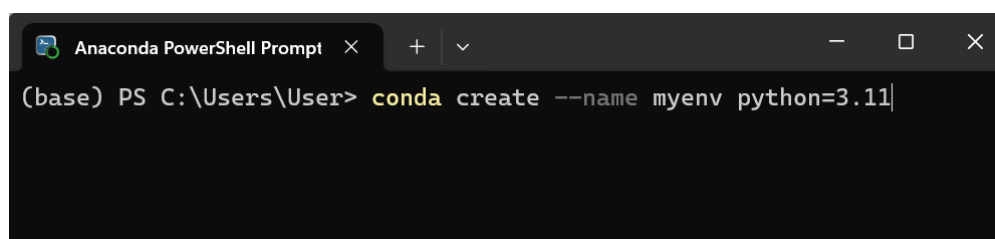
3. Install Miniconda

Miniconda is the recommended approach for installing TensorFlow with GPU support. It creates a separate environment to avoid changing any installed software in your system. This is also the easiest way to install the required software especially for the GPU setup.

Download the [Miniconda Windows Installer](#). Double-click the downloaded file and follow the instructions on the screen.

3. Create a virtual environment

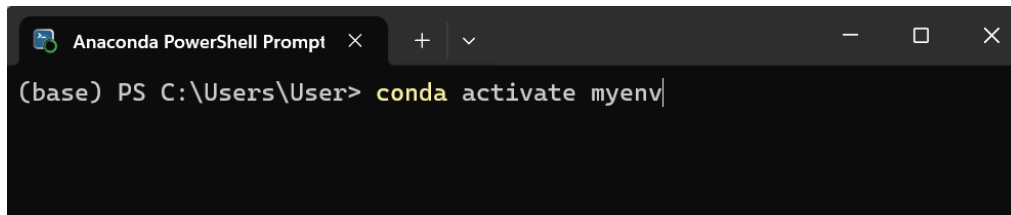
- ✓ Open Anaconda Powershell Prompt
- ✓ `conda create --name myenv python=3.11`



```
Anaconda PowerShell Prompt
(base) PS C:\Users\User> conda create --name myenv python=3.11
```

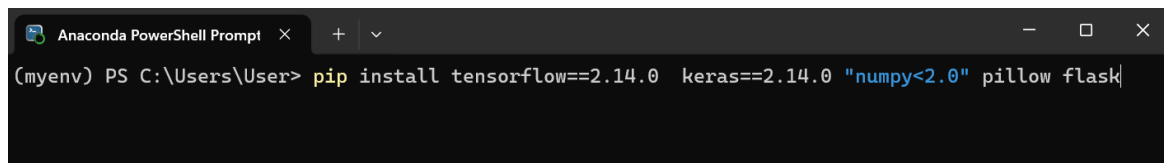
4. Install the required dependency:

- ✓ `conda activate myenv`



```
Anaconda PowerShell Prompt x + v - □ x
(base) PS C:\Users\User> conda activate myenv|
```

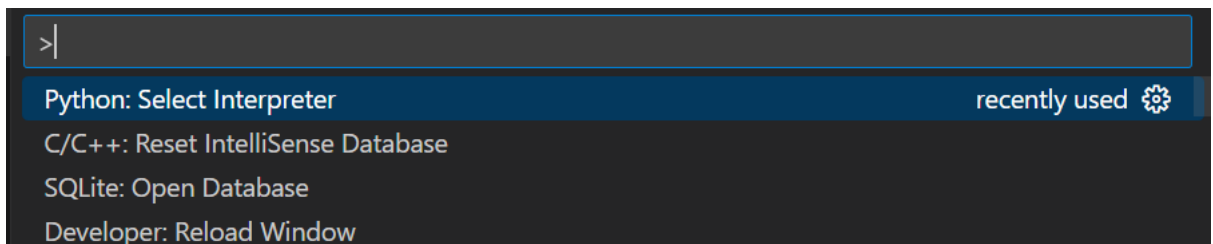
- ✓ `pip install tensorflow==2.14.0 keras==2.14.0 "numpy<2.0" pillow flask`



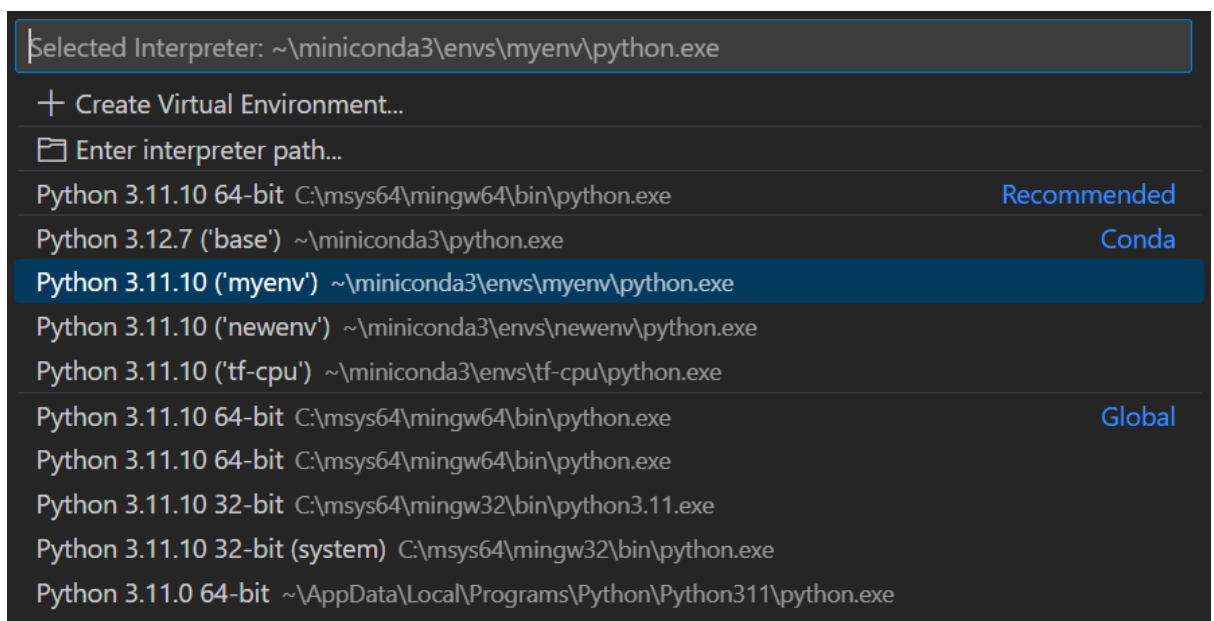
```
Anaconda PowerShell Prompt x + v - □ x
(myenv) PS C:\Users\User> pip install tensorflow==2.14.0 keras==2.14.0 "numpy<2.0" pillow flask|
```

5. Activate the virtual environment in VSCode for the respective folder

- ✓ Press “Ctrl + Shift + P”
- ✓ Select “Python Select Interpreter”

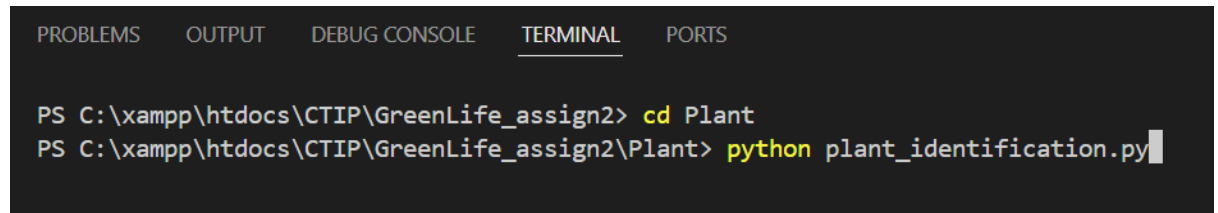


- ✓ Choose the created virtual environment (myenv)



6. Open terminal

- ✓ Direct to the file directory (`cd Plant`)
- ✓ Start the flask app (`python plant_identification.py`)



A screenshot of a terminal window with a dark background. At the top, there are five tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. Below the tabs, the terminal shows two lines of text. The first line is a PowerShell prompt 'PS C:\xampp\htdocs\CTIP\GreenLife_assign2>' followed by the command 'cd Plant'. The second line is a PowerShell prompt 'PS C:\xampp\htdocs\CTIP\GreenLife_assign2\Plant>' followed by the command 'python plant_identification.py'. A white cursor is visible at the end of the second line.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\xampp\htdocs\CTIP\GreenLife_assign2> cd Plant
PS C:\xampp\htdocs\CTIP\GreenLife_assign2\Plant> python plant_identification.py
```